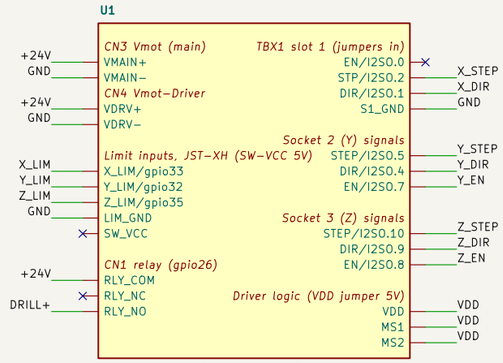
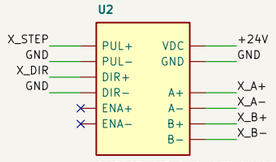


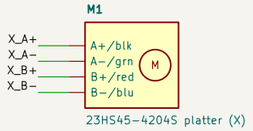
24 V 6 A brick + barrel-to-screw adapter



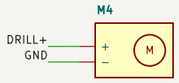
PiBot V4.96 Pro (ESP32, FluidNC)



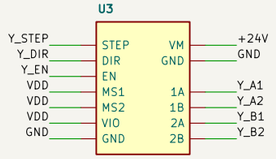
DM542T, DIP 1−8: OFF ON OFF OFF ON ON OFF ON



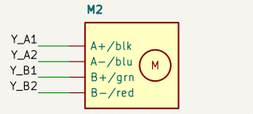
23HS45−4204S platter (X)



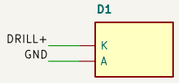
555 drill motor (24 V)



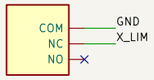
TMC2209 in socket 2 (Y), Vref 0.9−1.0 V



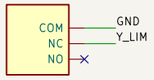
17HE08−1004S threader (Y)



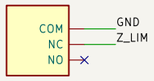
1N5822 flyback (recommended)



X index (nail 0)

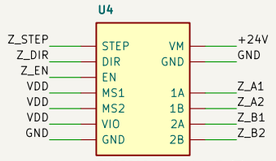


Y home (tip outside)

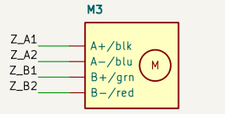


Z home (drill up)

NOTES
1. Pins follow the PiBot V4.96 Pro factory YAML and schematic (sch496, 2025−10−24), checked 2026−11−05, and ../fluidnc−config.yaml. Steppers are on the I2S shift register (I2SO.n).
2. Jumpers, 9 in total: VDD on 5V, SW−VCC on 5V, RLY (GPIO26) on relay, MS1 + MS2 on the VDD side (1/16), slot 1 routing x4. USB−5V, RST, SLP, NULL, SPI and slots 2−6 routing: none. Slot 1's StepStick socket stays EMPTY: a driver in a jumpered slot destroys it and the board.
3. Power, brick at 24.0 V max: CN3 "Vmot" (orange plug, main) and CN4 "Vmot−Driver" (sockets) both from the lever nuts, plus the DM542T VDC. "VMOT Main" on header J3 is an OUTPUT.
4. DM542T common cathode: STP to PUL+, DIR to DIR+, GND to PUL− and DIR−, ENA open. 5V/24V selector (V4.0) on 5V. DIP SW1−3 OFF ON OFF (2.36 A RMS), SW4 OFF, SW5−8 ON ON OFF ON (3200).
5. TBX1 has no pin labels. Pin 1−4 with routing jumpers in: EN, STP, DIR, GND. TBY2/TBZ3 pin 1−4 = 2B, 2A, 1A, 1B: each coil on two neighbouring screws.
6. TMC2209 standalone, 1/16: MS1 = MS2 = HIGH. Vref 0.9−1.0 V, max 1.4 V (needs CN4 powered).
7. Endstops COM to GND, NC to Signal: closed at rest, opens when hit or if a wire breaks; config pins have no :low. Input LEDs are lit at rest and go out when a switch is pressed.
8. Relay CN1: pin 1 COM, pin 2 NC (unused), pin 3 NO. D1 across the drill, stripe to +.
9. Small NEMA17 17HE08−1004S coils: black/blue = A, green/red = B. NEMA23: black/green, red/blue.



TMC2209 in socket 3 (Z), Vref 0.9−1.0 V



17HE08−1004S drill Z

Pins mirror machine/fluidnc−config.yaml
Generated by machine/wiring/make_schematic.py
crobison/string−art

Sheet: /
File: wiring.kicad_sch

Title: String art machine wiring

Size: A3	Date: 2026−11−04	**Rev: 2**
KiCad E.D.A. 10.0.8		Id: 1/1